Case Study 9: Simple Password Length Validator

1. Problem Statement

Read a sample password and check whether it contains at least 8 characters. Display a suitable message.

2. Objectives

1. Read a password string.
2. Use the len() function to count characters.
3. Check the minimum length using a conditional statement.
4. Display the validation result.

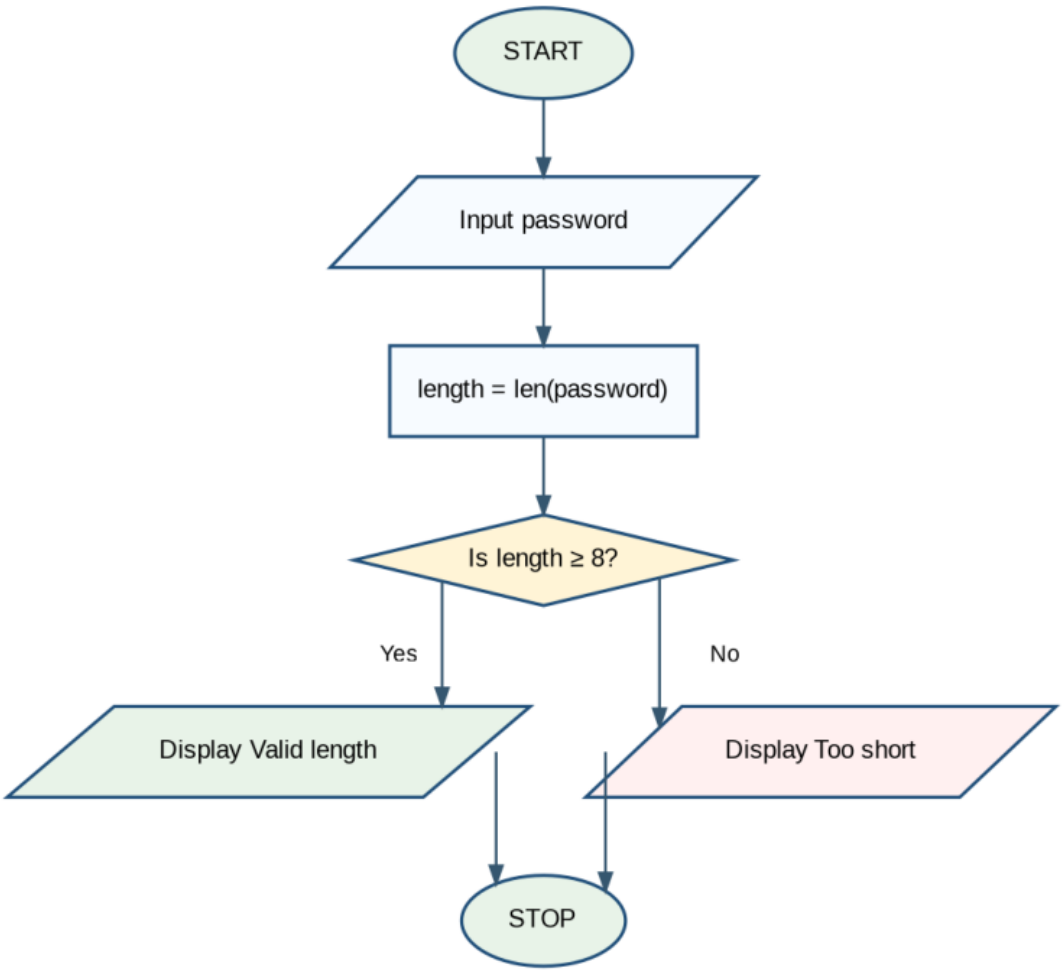
3. Assumptions

1. Minimum password length is 8 characters.
2. This is only a length check and does not test password strength.

4. Algorithm

1. Start.
2. Read the password.
3. Find its length using len().
4. If length is at least 8, display Valid length.
5. Otherwise display Too short.
6. Stop.

5. Flowchart



6. Pseudocode

```
INPUT password
length = LEN(password)
IF length >= 8 THEN
    DISPLAY "Valid length"
ELSE
    DISPLAY "Too short"
```

7. Complete Python Source Code

```
password = input("Enter a sample password: ")
length = len(password)
print("Number of characters:", length)
if length >= 8:
    print("Password length is valid")
else:
    print("Password is too short")
```

8. Test Case and Expected Output

Input = campus123.

Length = 9.

Expected output: Password length is valid.

9. Conclusion

The program uses string input, len(), and a conditional statement to validate password length.